ConMamba: CNN and SSM High-Performance Hybrid Network for Remote Sensing Change Detection

Zhiwei Dong, Genji Yuan, Zhen Hua, and Jinjiang Li

***Abstract*— Accurate remote sensing change detection (RSCD) tasks rely on comprehensively processing multiscale information from local details to effectively integrate global dependencies. Hybrid models based on convolutional neural networks (CNNs) and Transformers have become mainstream approaches in RSCD due to their complementary advantages in local feature extraction and long-term dependency modeling. However, the Transformer faces application bottlenecks due to the high secondary complexity of its attention mechanism. In recent years, state-space models (SSMs) with efficient hardware-aware design, represented by Mamba, have gained widespread attention for their excellent performance in long-series modeling and have demonstrated significant advantages in terms of improved accuracy, reduced memory consumption, and reduced computational cost. Based on the high match between the efficiency of SSM in long sequence data processing and the requirements of the RSCD task, this study explores the potential of its application in the RSCD task. However, relying on SSM alone is insufficient in recognizing fine-grained features in remote sensing images. To this end, we propose a novel hybrid architecture, ConMamba, which constructs a high-performance hybrid encoder (CS-Hybridizer) by realizing the deep integration of the CNN and SSM through the feature interaction module (FIM). In addition, we introduce the spatial integration module (SIM) in the feature reconstruction stage to further enhance the model's ability to integrate complex contextual information. Extensive experimental results on three publicly available RSCD datasets show that ConMamba significantly outperforms existing techniques in several performance metrics, validating the effectiveness and foresight of the hybrid architecture based on the CNN and SSM in RSCD.**

***Index Terms*— Mamba, remote sensing change detection (RSCD), state space models (SSMs), Transformers.**

I. Introduction

REMOTE sensing change detection (RSCD) is a technology that employs state-of-the-art image processing and pattern recognition algorithms to monitor surface changes. Through fine comparison of high-resolution remote sensing images acquired in different time series, this technology can accurately track and identify change events in a specified area. This capability is critical to unraveling the mechanisms of surface dynamics. Given its wide applicability in multiple fields, RSCD is of indispensable value in key areas such as urban development planning, disaster impact assessment, forest resource monitoring, environmental protection strategies, and agricultural trend analysis.

Led by rapid advances in deep learning technologies, the RSCD field is undergoing a fundamental shift from relying on traditional manual feature extraction to automated feature recognition using deep learning algorithms. This shift has been marked by the introduction of convolutional neural networks (CNNs) [1] and Visual Transformers (ViTs) [2], which iconized the application frontiers of deep learning. Despite their respective significant advantages, both algorithms expose limitations when dealing with specific scenarios. In particular, due to the local receptive field limitation of CNNs, their lack of capability in long-range feature correlation capture has guided researchers to explore strategies such as combining position-channel attention (e.g., MACGGNet [3]) and fusion of channel and spatial attention (e.g., IFNet [4] and DTCDSCN [5]), to enhance the performance of the models for spectral and spatial information processing. However, these CNN models enhanced by attention mechanisms [6], [7], [8], [9], [10], [11] are still deficient in global feature representation. In contrast, the Transformer model [12], [13], [14], [15], [16] relies on its self-attention mechanism to effectively address the global dependency modeling challenge and achieve optimization of the network architecture.

Although the Transformer architecture excels in global feature capture, its shortcomings in local feature extraction and detail preservation reveal its potential limitations. In the face of these challenges, innovative architectures that combine the strengths of the CNN and Transformer, such as ICIF-Net [17], achieve deep fusion of feature information by introducing a two-branch multilevel interaction network, which brings significant performance enhancement to the RSCD task. However, this enhancement relies on the Transformer's self-attention mechanism, leading to a significant increase in computational cost when processing large-scale image data.

In recent years, the application of state-space models (SSMs) [18] in several fields has made significant progress,

Received 25 April 2024; revised 18 August 2024 and 15 October 2024; accepted 7 November 2024. Date of publication 11 November 2024; date of current version 20 November 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 61772319, Grant 62002200, Grant 62202268, and Grant 62272281; in part by Shandong Natural Science Foundation of China under Grant ZR2023MF026 and Grant ZR2021QF134; and in part by Yantai Science and Technology Innovation Development Plan under Grant 2022JCYJ031. *(Corresponding author: Genji Yuan.)*

Zhiwei Dong and Zhen Hua are with the School of Information and Electronic Engineering, Shandong Technology and Business University, Yantai 264005, China.

Genji Yuan and Jinjiang Li are with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai 264005, China (e-mail: yuangenji@outlook.com).

Digital Object Identifier 10.1109/TGRS.2024.3495216

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

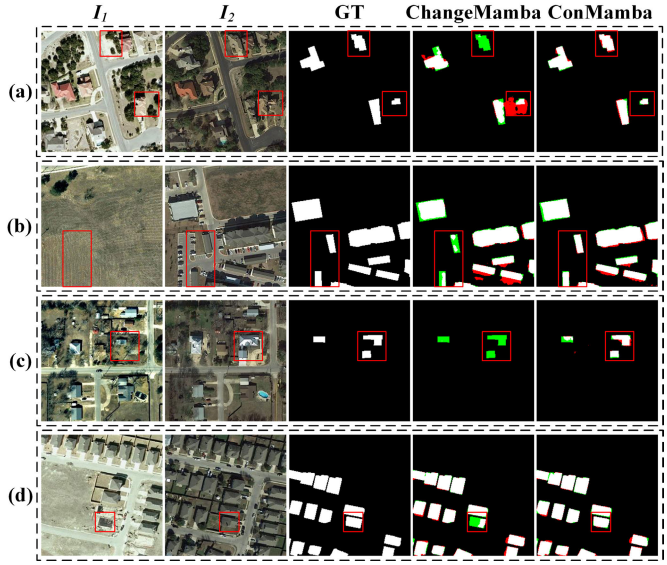


Fig. 1. Visualization results of ConMamba's detection of changes in tiny building targets in remotely sensed images. The figure is divided into five columns: the first and second columns are multitemporal images; the third column is ground-truth labeling (GT); the fourth column is the prediction results of ChangeMamba; and the fifth column is the prediction results of ConMamba. By comparing (a) and (b), it can be found that ConMamba can be seen to have excellent global feature extraction capability while compensating for ChangeMamba's shortcomings in local area CD. (c) and (d) Robustness of ConMamba to local changes in complex building environments, which can effectively deal with environmental interference and accurately recognize subtle features.

especially the introduction of the Mamba model [19], which is similar to the Transformer architecture and can effectively capture long-distance dependencies, but its computational complexity grows linearly with the input scale, which significantly improves the computational efficiency. Based on this feature, ChangeMamba [20] proposes a spatiotemporal modeling mechanism for three types of change detection (CD) tasks, which facilitates the full interaction of spatiotemporal features of multitemporal images by efficiently combining with the Mamba model to achieve accurate CD. However, a single Mamba architecture [21], [22], [23], [24], [25] with a global sense-field-first design struggles to adequately capture local information in intensive prediction tasks (e.g., CD), which imposes certain limitations on task performance.

To address the above problems, this article proposes an efficient CNN and SSM fusion framework, aiming to fully utilize the advantages of both, that is, capturing long-range dependent information by SSM and accurately extracting local details by the CNN. On this basis, we propose for the first time the ConMamba architecture, a hybrid model combining the CNN with state-of-the-art SSM techniques, aiming at exploring its potential application in RSCD tasks (see Fig. 1).

The core design of ConMamba is an efficient hybrid encoder, CS-Hybridizer, which realizes the efficient collaborative work between the CNN and SSM through the introduction of a feature interaction module (FIM) and achieves seamless fusion from local details to global features. Compared with the existing hybrid model of the CNN and Transformer, the CS-Hybridizer not only effectively circumvents the higher-order computational complexity problem of the hybrid model, but also makes up for the shortcomings of the standalone Mamba architecture in local information extraction. In addition, ConMamba introduces residual dilation convolution in the decoder part and constructs a spatial integration module (SIM), which further enhances the integration and expression of contextual information. Thanks to this innovative network design, the experimental results of ConMamba on several public datasets show that it outperforms the existing state-of-the-art technical solutions in several key performance indicators, which fully validates its excellent performance and application prospects in the field of RSCD.

Our main contributions can be summarized as follows.

1) For the first time, we explore the potential of a hybrid model of the CNN and SSM in the field of RSCD. Through extensive experimental validation on three publicly available RSCD datasets, the ConMamba model shows significant competitive advantages in several key performance metrics, which fully demonstrates the effectiveness and promise of the model in this field.
2) An efficient hybrid encoder, called CS-Hybridizer, is designed and proposed. CS-Hybridizer successfully integrates the CNN and SSM by introducing a simple FIM, which can capture both global information and local features, thus optimizing the extraction of multilevel remote sensing image features. This encoder not only circumvents the high subcomplexity problem caused by the self-attention mechanism, but also makes up for the shortcomings of the standalone Mamba architecture in local information extraction, and is expected to become a widely used and optimized general-purpose encoder in RSCD.
3) The SIM is introduced as a decoder for ConMamba, which substantially improves the integration of contextual information. SIM further improves the model's ability to parse the change information by enhancing the interaction of spatiotemporal features, which enhances the performance of the overall network architecture.

II. Related Work

A. Visual Modeling in the Field of Remote Sensing Transformation Detection

In the research field of vision tasks, CNN and ViT architectures have become dominant techniques in recent years. CD tasks have historically relied on traditional machine learning methods such as support vector machines (SVMs) [26] and random forests [27], which are limited in the face of complex data and require manual feature extraction. The rise of deep learning has led to CNN-based automatic feature learning methods to effectively handle data diversity and complexity in CD, among which the U-Net-based [28] fully convolutional twin network models (FC-EF, FC-Siam-conc, and FC-Siam-diff) proposed by Daudt et al. [29] marked the initial contribution of deep learning in CD research. Zhang et al. [21] innovated models with better performance and faster convergence by integrating the attention mechanism in FC-Siam-conc [29]. Meanwhile, STANet [30] incorporates an adaptive attention mechanism to focus on key change regions through metric

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

learning with multiscale image segmentation. Aiming at the problem of spatial information loss due to continuous downsampling, Fang et al. [31]proposed the SNUNet-CD model based on NestedUNet [32], which optimizes the blurring problem of edge pixel recognition by preserving high resolution and details through dense hopping concatenation in codecs.

In the field of CD, traditional CNNs are mainly optimized for local feature extraction, ignoring the CD task's need for overall scene contextual understanding, that is, the global information processing capability necessary to distinguish real changes from background noise. In contrast, ViT, through its self-attention mechanism, can capture long-range intraimage dependencies, enabling an in-depth understanding of the interpixel interactions of the entire image scene, which is crucial for the CD task. Based on this principle, SwinSUNet [33], ChangeFormer [15], and ScratchFormer [34] adopt the Transformer structure and design a twin U-shaped architecture to enhance the capture of the global field of view. However, the performance of the Transformer model is limited in some scenarios due to the lack of effective inductive bias, prompting some studies to explore the parallel use of the CNN and Transformer [35], [36], [37], [38], [39] to compensate for their respective shortcomings.

The DBFGAN model [40] achieves an effective fusion of location information and global semantic information through the interlayer complementary mechanism between the CNN and Transformer, while the FTN model [41] adopts a sequential integration strategy, which cleverly combines the local feature extraction advantages of the CNN with the superior performance of Transformer in modeling global dependencies. Nevertheless, this hybrid architecture faces performance bottlenecks due to the inherent secondary computational complexity of the Transformer.

Recent studies have revealed that SSMs, represented by Mamba, exhibit high efficiency in modeling long-distance interactions and maintain linear computational complexity. Based on this finding, this study is dedicated to efficiently integrating the CNN with SSM to improve modeling efficiency while maintaining model performance.

B. State-Space Models

SSM provides a framework for transforming sequence data and has shown the ability to handle long-term dependencies, but is difficult to train due to its high computational and memory usage requirements. Mamba has not only simplified the architecture of SSM by using custom parameterization techniques and algorithms optimized for hardware but also dramatically improved its efficiency in dealing with long-distance dependencies between language and genome sequences. With the success of SSM in several domains, its use in computer vision tasks has grown dramatically. In particular, S4ND [42] first applied SSM modules to the vision domain, deepening the understanding of modeling visual data in all dimensions. Vmamba [43] subsequently successfully tackled the orientation perception problem triggered by the difference between 1-D sequences and multichannel images by developing Mamba-based visual architectures and cross-scan modules. Recent studies, such as Vm-unet [44], U-Mamba [45], and SegMamba [46], have further optimized and adapted Mamba-based models for medical image segmentation tasks. In addition, the application of Mamba models in remote sensing has also received extensive attention. ChangeMamba [20] introduces the Mamba model to the remote sensing image CD task for the first time, demonstrating the efficient capability and adaptability of the Mamba model in analyzing the changes of time-series remote sensing data. RSMamba [47]) is used in the remote sensing image classification process by combining shuffle and forward and backward scanning, it successfully improves the accuracy of remote sensing image classification. Although the above model fully exploits the unique advantages of SSM in capturing long-term temporal dependencies, it ignores the limitations of SSM in capturing local details. Based on this, this article attempts to integrate the respective advantages of CNN and SSM layers in local and global feature extraction to compensate for the shortcomings of SSM when used alone.

III. Methodology

A. Architecture Overview

Fig. 2 illustrates the general architecture of our ConMamba. The architecture admits a set of temporally sequenced images as inputs (prechange image and postchange image) and is composed based on three core technical components: a high-performance hybrid encoder (CS-Hybridizer) that combines the strengths of the CNN and SSM, a difference module (DM) that emphasizes the difference between the images before and after the change, and a decoder SIM that predicts the binary change map.

In ConMamba's workflow, image pairs are first passed through a parallel configuration of CS-Hybridizer encoders, which share weights, to perform feature extraction on the time-series images separately, producing feature representations of three different stages (F_{pre_i} and F_{pos_i}). These features were then fed into the DM module, where an enhanced feature fusion operation was performed to emphasize the change signals while suppressing the background noise and ensuring that the model focused on the key changes. The DM-processed features were then passed to the SIM decoder, where efficient contextual information fusion was achieved through residual dilation convolution, enhancing the model's semantic understanding of the change region. In the final stage of the process, the synthesized fused features are fed into the mask prediction layer, which ultimately generates accurate binary change masks. The exact workflow of ConMamba is shown in Algorithm 1.

B. CS-Hybridizer Encoder

Powerful feature extraction capability becomes necessary in processing high-resolution remote sensing images with high dimensionality and complex textures. Although CNNs are effective in parsing complex image structures, fine-grained capture of global information requires a wider sensory field with optimized global modeling. To bridge this gap, we explore a strategy of incorporating the SSM branch at

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

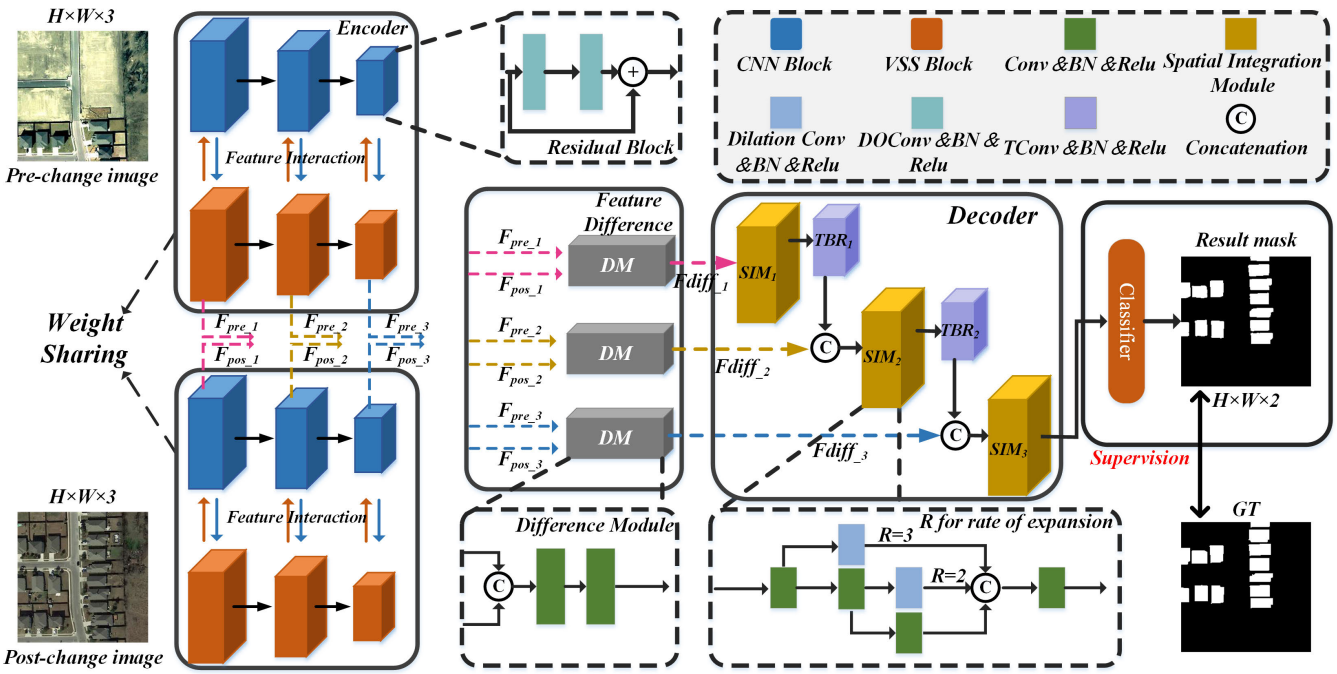


Fig. 2. Schematic of ConMamba's network structure. The gray area in the upper right corner is the diagram annotation.

Algorithm 1 Implementation Process of Our ConMamba Model

Input: I_1, I_2 (*bitemporal image*))
Output: M(*a prediction change mask*)
// *step*1 : *CS-Hybridizer feature extraction*
for i *in* $\{1, 2, 3\}$ **do**
  $\mathbf{F_{pre_i}}, \mathbf{F_{pos_i}} = CS\text{-}Hybridizer(\mathbf{I_1}, \mathbf{I_2})$
end
// *step*2 : *Aggregation of differential features*
for i *in* $\{1, 2, 3\}$ **do**
  $\mathbf{F_{diff_i}} = DM(\mathbf{F_{pre_i}}, \mathbf{F_{pos_i}})$
end
// *step*3 : *Information integration*
for i *in* $\{1, 2, 3\}$ **do**
  if $i == 1$ **then**
    $\mathbf{F^i_{sim}} = TBR_i\left(SIM_i\left(\mathbf{F_{diff_i}}\right)\right)$
  if $i == 2$ **then**
    $\mathbf{F^i_{sim}} = TBR_i\left(SIM_i\left(CAT\left(\mathbf{F_{diff_i}}, \mathbf{F^{i-1}_{sim}}\right)\right)\right)$
  else
    $\mathbf{F^i_{sim}} = SIM_i\left(CAT\left(\mathbf{F_{diff_i}}, \mathbf{F^{i-1}_{sim}}\right)\right)$
  end if
end for
// *step*4 : *Classification head*
$\mathbf{M} = CH(\mathbf{F^3_{sim}})$

the encoding stage. SSM demonstrates long-range dependency capture capabilities that match those of the Transformer while providing superior computational and parametric efficiencies. This technique complements the capabilities of the CNN in local feature analysis. Based on this, we developed the CS-Hybridizer hybrid encoder by fusing the CNN and SSM branches through FIM. The encoder is designed to comprehensively and efficiently capture and process complex features in high-resolution images, optimizing the ability to integrate global and local information for processing.

Fig. 3 shows the details of the design of the CS-Hybridizer architecture, with a special emphasis on the interaction mechanism between the CNN and the SSM branches. In the initialization phase, the input features are processed through the First Block (a convolutional layer containing different convolutional kernel sizes) to achieve channel dimensionality adjustment (changing a three-channel image into a 64-channel image).

The tuned features are then fed in parallel to the CNN branch for local feature extraction and the SSM branch for global information capture. In the CNN branch, the features are processed by Residual Block to enhance the depth of the model's representation of the input data, followed by further channel and spatial dimension optimization through convolutional layers and downsampling operations in FIM.

The optimized features are concatenated with the initial inputs of the SSM branch and the concatenated features are passed through the VSS Block for global feature capture, this step is critical as it utilizes the long-range dependency modeling capability of the SSM branch to enhance the understanding of wide areas in the remote sensing image. Afterward, the features are passed again to the CNN branch through the convolutional layers and upsampling operations in the FIM to further refine and fine-tune the features.

This iterative and cyclic simple interaction mode not only fully grasps the local details and global information but also promotes the effective exchange and fusion of in-depth features through continuous information feedback loops, which improves the performance and accuracy of the overall network

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

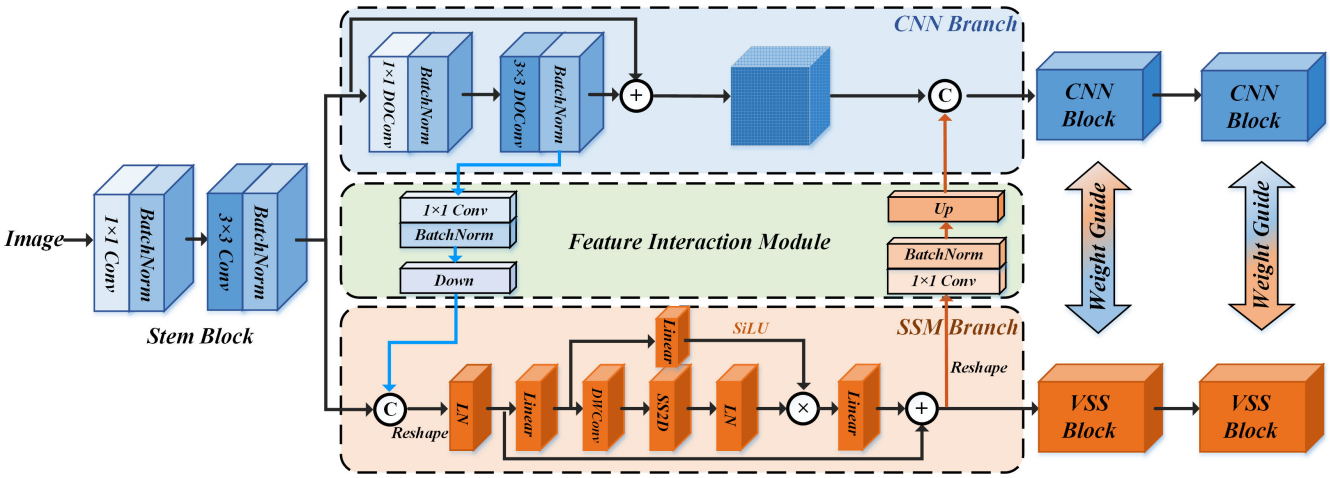


Fig. 3. Schematic of the network structure of CS-Hybridizer encoder. The top blue region represents the CNN branch, the middle region represents the FIM, and the bottom orange region is the SSM branch.

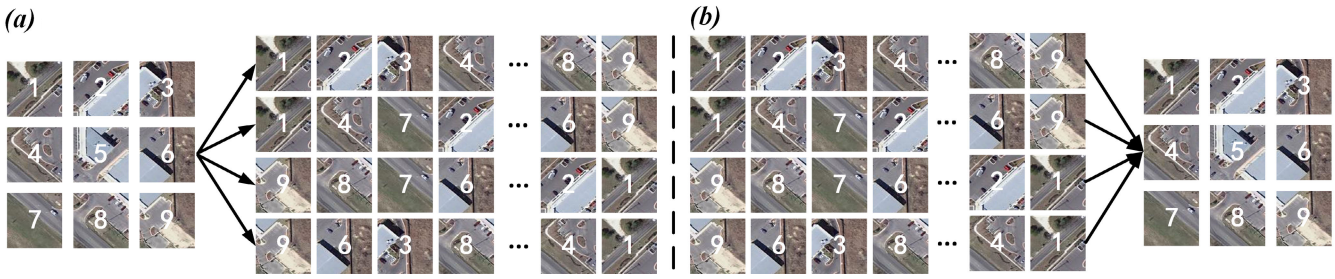


Fig. 4. (a) Scan expanding operation in SS2D. (b) Scan merging operation in SS2D.

in the RSCD task. In addition, this easy interaction mode ensures an objective assessment of the hybrid CNN- and SSM-based model and avoids the introduction of additional confounding factors due to the complex combination approach, which in turn allows for a more accurate assessment of the applicability and potential value of this hybrid model in the field of RSCD. Therefore, we adopt this combination approach to systematically analyze and evaluate the characteristics and performance of the model itself.

The specific channel and spatial dimension changes are shown in Fig. 3. Assuming that the input features f are given, the mathematical description of the process is as follows:

$$f_s = \mathrm{SB}(f) \tag{1}$$

$$F = \mathrm{FIM}(\mathrm{CNN}(f_s), \mathrm{SSM}(f_s)) \tag{2}$$

where SB denotes the Stem Block and F is the feature after FIM interaction.

1) SSM Branch: In the RSCD task, it is crucial to accurately model global features with multiscale features. Despite the ability of the Transformer architecture to extract global information, it imposes a significant increase in computational cost when faced with excessively long feature sequences. For this reason, the SSM architecture is designed to reduce the complexity of the attention mechanism for modeling long sequences from quadratic to linear, achieving high accuracy, low computational cost, and memory efficiency in vision tasks. In addition, due to its excellent ability to model time series data, SSM is particularly well suited to deal with dynamic changes in remotely sensed images, making SSM ideal for processing such data.

Based on the above advantages, we introduced a Mamba-based visual state space (VSS) module at the Con-Mamba encoder stage as the core unit of the SSM branch. The VSS module employs a 2-D selective scanning (SS2D) strategy to accommodate the specific processing needs of 2-D image data. As shown in Fig. 4, the SS2D technique generates four mutually independent feature sequences by unfolding image patches along four independent directions, which are subsequently processed by SSM. This strategy ensures that the features captured from different viewpoints are considered together, and finally, a complete 2-D feature mapping is generated by the fusion operation. Given the input feature f, the mathematical representation of the output feature $\overline{f}$ of SS2D is

$$\overline{f}_v = S6(\mathrm{expand}(f, v)) \tag{3}$$

$$\overline{f} = \mathrm{merge}\left(\overline{f}_1, \overline{f}_2, \overline{f}_3, \overline{f}_4\right) \tag{4}$$

where $v \in V = \{1, 2, 3, 4\}$ is the four different scan directions. expand$(\cdot)$ and merge$(\cdot)$ correspond to the scan unfolding and scan merging operations, respectively. The selective scanning spatial state sequence model (S6) in 4 is the core SSM operator of the VSS module. It allows each element in the 1-D array to interact with any previously scanned samples by compressing the hidden states.

The detailed construction of the VSS block is shown in Fig. 3. Assuming that the input feature is f, the corresponding

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

mathematical representation of its operation flow is

$$f_1 = \mathrm{LN}(f) \tag{5}$$

$$f_1' = \mathrm{LN}(\mathrm{SS2D}(\circ(\mathrm{Linear}(f_1)))) \tag{6}$$

$$f_1'' = \mathrm{SiLU}(\mathrm{Linear}(\mathrm{Linear}(f_1))) \tag{7}$$

$$F = \mathrm{Linear}\left(f_1' \cdot f_1''\right) + f_1 \tag{8}$$

where $\circ$ denotes deep convolution, LN denotes LayerNorm, $\cdot$ denotes multiplication operation, and F denotes the final output result.

2) CNN Branch: In the design stage of the CNN Branch, we use the residual structure to optimize the convolutional layer. Furthermore, to make the CNN Branch more computationally efficient, we replace all the conventional convolutions in the residual block with DO-Conv [48].

DO-Conv is an innovative technique that integrates deep convolution with traditional convolution with trainable kernels. Where the trainable kernels in deep convolution and traditional convolution are labeled as $D \in \mathbb{R}^{(M\times N)\times D_{\mathrm{mul}}\times C_{\mathrm{in}}}$ and $W \in \mathbb{R}^{C_{\mathrm{out}}\times D_{\mathrm{mul}}\times C_{\mathrm{in}}}$, respectively, satisfying $D_{\mathrm{mul}} = M \times N$, where M and N denotes the spatial dimensions of the convolutional layer patches, C_{in} denotes the number of input feature channels, C_{out} denotes the number of output channels, D_{mul} denotes the depth multiplier, and the value of D_{mul} is fixed in the application of DO-Conv. For the input patch $P \in \mathbb{R}^{(M\times N)\times C_{\mathrm{in}}}$, DO-Conv produces a feature output of C_{out} similar to conventional convolution. In addition, DO-Conv introduces two operational mechanisms, feature combination and kernel combination. The computationally more efficient kernel combination approach is adopted in the study, using the symbol $\circledast$ to denote the DO-Conv convolution operation, then the kernel combination process of DO-Conv can be represented as

$$O = (D, W) \circledast P = D^T \circ W * P \tag{9}$$

where $*$ and $\circ$ denote conventional and deep convolution operations, respectively, and $D^T \in \mathbb{R}^{D_{\mathrm{mul}}\times(M\times N)\times C_{\mathrm{in}}}$ is the transpose of kernel D about its first two axes. The kernel combination is implemented using $W' \in \mathbb{R}^{C_{\mathrm{in}}\times(M\times N)\times C_{\mathrm{Out}}}$ to integrate deep and conventional convolution, which ensures that the sensory field is $M \times N$ throughout DO-Conv.

Although DO-Conv slightly increases the number of parameters compared to traditional convolution, it improves computational efficiency due to its independent depth and point-by-point convolution operations. This separate convolutional approach improves the interpretability of the model and helps to understand its operation mechanism in depth. As a result, DO-Conv's combined advantages of computational efficiency, generalization, and interpretability enable the CNN branch to achieve improved performance under the constraints of limited computational resources.

3) Feature Interaction Module: Given the significant differences between CNNs and SSMs regarding feature representation capability and hierarchical structure, direct integration of their features makes it difficult to bridge the gap. It fails to fully utilize their complementary properties. To address this challenge, we adopt an FIM strategy to build an efficient encoder architecture, CS-Hybridizer, by integrating CNN and SSM branch features layer by layer (see Fig. 3). In the initial stage, the intermediate features of the CNN branch are size-matched by 1×1 convolution and downsampling operations to fit the feature scales of the SSM branch and enable subsequent cross-branch interactions. After the first interaction, the features enter the VSS module for global feature modeling, are then dimensionally reduced by 1×1 convolution and upsampling operations, and then fed back to the CNN branch. With this simplified step-by-step interaction strategy, we achieve an effective integration of CNN and SSM features, which not only bridges the representation hierarchy but also significantly enhances the synergy between the two at different feature dimensions, thus improving the adaptability of the CS-Hybridizer encoder to complex data. In addition, the adoption of this strategy helps to control the variables to ensure an objective assessment of the performance of the hybrid CNN and SSM models so that their potential advantages in RSCD can be fully demonstrated.

C. Difference Module

To accurately quantify and analyze the multilayer feature differences between the feature images output by the parallel CS-Hybridizer encoder before and after the change, we deployed the DM. These modules include two sizes of Conv2D layers (3 and 1), the ReLU activation function, and the batch normalization layer. The mathematical representation of the DM is as follows:

$$F_{\mathrm{diff}_i} = \mathrm{CRB}_1\left(\mathrm{CRB}_3\left(\mathrm{Cat}\left(F_{\mathrm{pre}_i}, F_{\mathrm{pos}_i}\right)\right)\right). \tag{10}$$

In this expression, CRB stands for the composite operation of Conv2D, ReLU, and BatchNorm, and the subscript refers to the size of the convolutional kernel. F_{pre_i} and F_{pos_i} refer to the feature maps of the pre- and postchange images in the ith layer, respectively, F_{diff_i} denotes the difference of the pre- and postchange features in the ith layer, and Cat denotes the stitching operation. This DM significantly improves the accuracy of CD by learning the optimal distance metric at each level instead of only calculating the absolute difference between F_{pre_i} and F_{pos_i}.

D. SIM Decoder

As shown in Fig. 2, we employ SIM as the core unit in the decoder stage. SIM consists of three layers that process the output of differential features from different CD stages and integrate these features through an interactive learning mechanism to infer the final masked image. The mathematical representation of this process is given as

$$F_{\mathrm{sim}}^i = \begin{cases} \mathrm{TConv}_i\left(\mathrm{SIM}_i\left(F_{\mathrm{diff}_i}\right)\right), & i = 1 \\ \mathrm{TConv}_i\left(\mathrm{SIM}_i\left(\mathrm{Cat}\left(F_{\mathrm{diff}_i}, F_{\mathrm{sim}}^{i-1}\right)\right)\right), & i = 2 \\ \mathrm{SIM}_i\left(\mathrm{Cat}\left(F_{\mathrm{diff}_i}, F_{\mathrm{sim}}^{i-1}\right)\right), & i = 3 \end{cases} \tag{11}$$

where TConv refers to the joint operation of transpose convolution, BatchNorm, and ReLU, and the subscript i indicates the layer level. F_{diff_i} and F_{sim}^i refer to the differential feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

inputs with corresponding feature outputs received by the SIM at layer i, respectively.

During SIM processing, the disparity features are first passed through a convolutional layer with a convolutional kernel size of 3 for initial channel dimension adjustment. The adjusted features are further processed along three separate paths: the first path applies an expansion convolution with an expansion rate of 3; the second path goes through an expansion convolution with an expansion rate of 2, followed by a convolutional layer with a convolutional kernel size of 3; and the third path goes directly through a convolutional layer with a convolutional kernel size of 3. The outputs of these three paths are subsequently fused with the cascaded convolutional layers to achieve feature aggregation. The flow of operations for each layer of SIM can be described by the following mathematical expression:

$$F'_{\text{diff_}i} = \text{CBR}_3\left(F_{\text{diff_}i}\right) \tag{12}$$

$$F^1_{\text{diff_}i} = \text{dc}_3\left(F'_{\text{diff_}i}\right) \tag{13}$$

$$F^2_{\text{diff_}i} = \text{dc}_2\left(\text{CBR}_3\left(F'_{\text{diff_}i}\right)\right) \tag{14}$$

$$F^3_{\text{diff_}i} = \text{CBR}_3\left(\text{CBR}_3\left(F'_{\text{diff_}i}\right)\right) \tag{15}$$

$$F^i_{\text{sim}} = \text{CBR}_3\left(\text{Cat}\left(F^1_{\text{diff_}i}, F^2_{\text{diff_}i}, F^3_{\text{diff_}i}\right)\right) \tag{16}$$

where CBR denotes the joint operation of normal convolution, BatchNorm, and ReLU, and the subscript denotes the convolution kernel size. DC denotes the joint operation of dilated convolution, BatchNorm, and ReLU, and the subscript denotes the dilatation rate. $F^j_{\text{diff_}i}$, $j \in (1, 2, 3)$ denotes the three intermediate outputs within the SIM, respectively.

E. Other Network Details

1) Classification Header: In the final stage of the decoder, the $H \times W \times 2$ segmentation mask is finally obtained by performing two convolution operations of 3×3 size for channel number conversion.

2) Loss Function: In terms of the loss function, we used the cross-entropy loss function [49] to optimize the model parameters. The loss function mathematical expression is

$$L = \frac{1}{H_0 \times W_0} \sum_{h=1,w=1}^{H,W} l(P_{hw}, Y_{hw}) \tag{17}$$

where H_0 and W_0 denote the height and width of the input image, respectively, $l(P_{hw}, y) = -\log(P_{hwy})$ is the cross-entropy loss, and P_{hw} and Y_{hw} are the predicted probability and labeling of the pixel at position (h, w), respectively.

IV. Experimental Results and Analysis

A. Experimental Settings

1) Experimental Parameter Settings: All experiments were conducted on an Ubuntu 18.04 system equipped with NVIDIA TITAN RTX GPUs, utilizing the PyTorch 1.11.0 environment to ensure sufficient computational resources. For model optimization, we implemented a stochastic gradient descent (SGD) strategy with momentum, where the momentum parameter was set to 0.99 and the weight decay was set to 0.0005. For both datasets, the initial learning rate was uniformly set to 0.01. In the experimental configuration of ConMamba, we set the maximum training period to 300 rounds. The same learning rate and number of training rounds were kept for all other comparison methods.

2) Evaluation Metrics: In this study, to address the extensive research background in the field of RSCD, we selected five core assessment metrics for performance evaluation, including Precision, Recall, $F1$, Mean Intersection and Union Ratio (mIoU), and Overall Accuracy (OA). These metrics were combined and applied to provide a comprehensive assessment of the efficacy of the ConMamba model. The mathematical definitions of the relevant evaluation metrics are described below

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{20}$$

$$\text{mIoU} = \frac{\text{TP}}{\text{FP} + \text{FN} + \text{TP}} \tag{21}$$

$$\text{OA} = \frac{\text{TP} + \text{pn}}{\text{TP} + \text{TN} + \text{FN} + \text{FP}} \tag{22}$$

where TP stands for true case, TN stands for true negative case, FP stands for false positive case, and FN stands for false negative case.

B. Description of the Datasets

LEVIR-CD [30] is a collection of optical imagery widely used for RSCD, provided by Google Earth. It includes 637 ultrahigh resolution (0.5 m/pixel) images, all sized at 1024×1024 pixels, covering residential, commercial, and industrial areas in several Texas cities. The dataset includes dual time-phase imagery that documents the movement of numerous areas and buildings. The seasonality of the image acquisition time and light variations pose challenges to the stability of the network performance. LEVIR-CD crops the images to 256×256 pixels by default and randomly divides them into a training set (7120 images), a validation set (1024 images), and a test set (2048 images) at a ratio of 7:1:2.

WHU-CD [50] is an optical dataset acquired from aerial equipment covering detailed information on numerous buildings in Christchurch, New Zealand. The ultrahigh-resolution images (0.075 m/pixel) of this dataset reveal a wealth of architectural detail, and the high resolution results in more prominent shadows around the buildings due to lighting effects. In the experiments, these environmental factors often resulted in biased target localization of the models. The original aerial image size of $32\,507 \times 15\,354$ exceeded the conventional memory processing capacity, so we nonoverlapping cropped the image to 256×256 size. Given the small amount of WHU-CD data, we divided the data in an 8:1:1 ratio to obtain 6096 training images, 762 validation images, and 762 test images.

GZ-CD [51] dataset is derived from Google Earth and records changes in the Guangzhou area between 2006 and 2019. This optical dataset contains a series of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

ultrahigh-resolution images ranging in size from 1006 × 1186 to 4936 × 5224, with a specific resolution of 0.55 m/pixel. The dataset covers a variety of building types, such as residential and industrial buildings, whose different colors increase the difficulty of target recognition. The images of GZ-CD are processed as 256 × 256 pixels without overlapping and are partitioned into training, validation, and test sets according to the ratio of 8:1:1, with the number of samples in the order of 2834, 400, and 325.

C. Comparison Experiments

1) Comparison With State-of-the-Art Methods: This section provides a systematic comparative analysis of ConMamba with the current mainstream methods, covering the following categories of methods: the CNN-based FC-Siam-Conc [29]; four methods based on attentional mechanisms, that is, IFNet [4], SUNet [31], DTCDSCN [5], and AERNet [52]; four methods combining the techniques of CNNs and Transformers, including BIT [53], ChangeFormer [15], FTN [41], and ICIF-Net [17]; and one approach based on the Mamba architecture, namely ChangeMamba [20]. This is followed by a brief overview of the rationale behind the above-compared approaches.

1) FC-Siam-Conc [29] is based on the U-Net architecture, where bi-phasic features are passed through a shallow network and fused by splicing, which in turn is integrated with deep semantic information.
2) IFNet [4] applies channel and spatial attention mechanisms to enhance the extraction of disparity features and optimizes the disparity feature extraction process through deep supervision methods.
3) SNUNet [31] relies on the UNet++ architecture to enhance the capture of details and semantics through enhanced hopping connections, while introducing an integrated channel attention module to process semantic information at different levels.
4) DTCDSCN [5] uses SE-ResNet as its infrastructure and integrates a dual-attention module containing both channel and spatial attention aimed at improving feature discrimination.
5) AERNet [52] introduces an edge refinement network that integrates multilevel contextual features through a Global Contextual Feature Aggregation Module (GCFAM). The model utilizes Attention Decoding Blocks (ADBs) and Edge Refinement Modules (ERMs) to enhance the identification and refinement of changing regions.
6) BIT [53] is constructed based on ResNet [54] and utilizes the Transformer technique to achieve long-range dependent processing of deep semantic information. The model strengthens the salient representation of detailed semantics by subtracting the different features and absolute value operation.
7) ChangeFormer [15] utilizes the Transformer architecture as its foundation and introduces an innovative DM that continuously integrates difference features across multiple layers to optimize the representation and processing of feature information.
8) FTN [41] utilizes ST for multiscale feature extraction, combined with pyramid structure and progressive attention module to optimize CD. In addition, the model performance is enhanced by accurate loss calculation for each layer output through deep supervision.
9) ICIF-Net [17] uses a dual-backbone architecture of the CNN and Transformer to extract features, but there is no direct interaction between them. To improve this limitation, a convolutional attention mechanism is introduced to optimize the fusion of same-resolution features.
10) ChangeMamba [20] explores for the first time the potential application of the Mamba architecture in the field of RSCD and proposes a spatiotemporal modeling mechanism applicable to three types of CD tasks. The method strengthens the spatiotemporal feature interaction of multitemporal images through efficient coupling with Mamba models, thus realizing high-precision CD.

2) Visualizing Comparative Analysis: In this section, we categorize and describe the experiments according to the dataset. All the experimental results are visually compared with the ground-truth (GT) data, where the red color indicates the misdetected area and the green color indicates the missed area.

a) LEVIR-CD: Fig. 5 shows a few selected samples of the LEVIR-CD dataset for visual comparative analysis. In the first set of samples, the chromatic aberration between dual-temporal images A and B due to seasonal changes, combined with the complex built environment, leads to varying degrees of leakage in CD for all comparison networks. Only ConMamba's results are largely consistent with GT, highlighting its superior ability to identify small-scale change targets. To verify ConMamba's ability to handle large-scale change targets, we selected samples (b)–(d) for detailed analysis, and demonstrated a local zoomed-in view on sample (b). Visual comparisons show that ConMamba exhibits excellent performance in all three sets of samples. In samples (b) and (d), the other comparison networks cause significant misclassification by failing to correctly handle shadows caused by illumination and color changes, mistaking shadows for changing targets. In addition, the trees in sample (c) are close in color to the surrounding buildings, which also leads to recognition errors. In contrast, ConMamba effectively excludes the effects of shadows and environmental disturbances and accurately identifies the true change targets. This achievement stems from ConMamba's in-house CS-Hybridizer hybrid encoder's ability to accurately handle long and short-distance dependencies and SIM's staged information processing strategy.

Table I demonstrates the superior performance of ConMamba on the LEVIR-CD dataset and validates its visualization with rigorous data. As shown in the table, ConMamba achieves the best performance on all four evaluation metrics, especially on the $F1$ value and mIoU metrics, where it improves 0.53% and 0.84% over the suboptimal network, and 1.67% and 2.85% compared to ChangeMamba, respectively. However, ConMamba does not reach the highest value in the Recall metric, which indicates a potential direction for further optimization of the model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

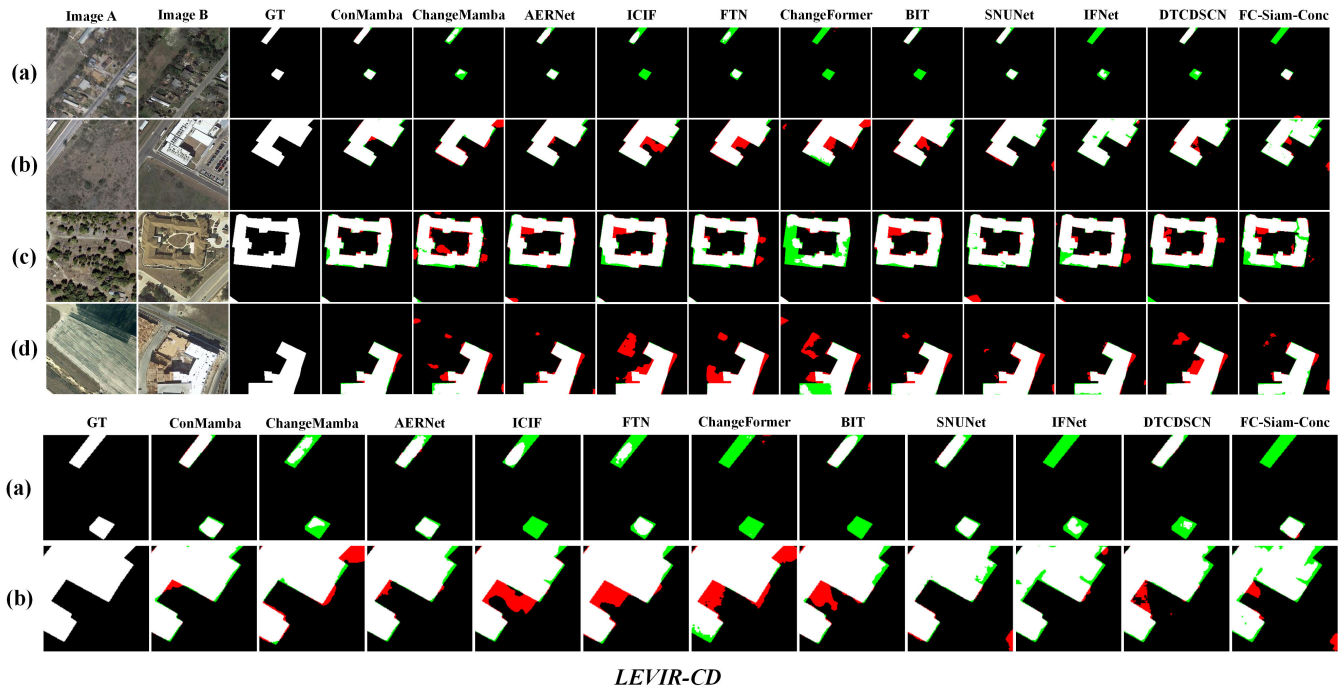


Fig. 5. Visualization results of ConMamba and comparison methods on the LEVIR-CD dataset. The upper images (a)–(d) display the overall detection results of dataset samples, while the lower images (a) and (b) provide zoomed-in views of key areas from the upper images (a) and (b).

TABLE I

RESULTS OF COMPARATIVE EXPERIMENTS ON THE LEVIR-CD DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUES HIGHLIGHTED IN RED AND THE SECOND HIGHEST IN BLUE

LEVIR-CD					
Model	Precision	Recall	F1	mIoU	OA
FC-Siam-Conc	86.37	88.15	87.25	77.39	98.69
DTCDSCN	88.53	86.83	87.67	78.05	98.77
IFNet	92.57	86.29	89.32	80.70	98.95
SNUNet	89.18	87.17	88.16	78.83	98.82
BIT	89.24	89.37	89.31	80.68	98.92
ChangeFormer	92.05	88.80	90.40	82.48	99.04
FTN	**92.71**	89.37	91.01	83.51	99.06
ICIF-Net	91.13	**90.57**	**91.18**	**83.85**	**99.12**
AERNet	89.97	**91.59**	90.78	83.11	99.07
ChangeMamba	91.32	88.64	90.04	81.84	98.99
ConMamba	**93.14**	90.06	**91.71**	**84.69**	**99.17**

b) WHU-CD: Fig. 6 shows the sample results of ConMamba and other comparative methods in dealing with complex building structures on the WHU-CD dataset. In sample (a), there are complex geometric building structures as well as visual disturbances caused by environmental factors. For example, in sample (a), the similar colors of the roof and the yard in the upper left corner lead to misidentification, while the small building in the lower left corner is usually ignored by the network due to its small size. To analyze the above complex regions in depth, we locally zoomed-in view on the relevant parts, and (a1) and (a2) shows the zoomed-in view upper-left and lower-left views, respectively. The analysis from (a1) shows that, except for ConMamba and FTN, the other comparison networks exhibit a lack of recognition accuracy when dealing with scenes with similar colors, exposing their limitations in distinguishing objects with close colors. In (a2), only ConMamba accurately recognizes the small building in the lower left corner, highlighting its superior ability in small targets and detail processing.

In sample (b), the protruding part in the center of the building is similar in color to the gray car on the road in Figure A. This color similarity may lead to misidentification in some networks. However, after local zooming and careful analysis of sample (b), ConMamba demonstrates its excellent recognition ability by accurately distinguishing and identifying real building variations. In samples c and d, the color similarity between roads and buildings poses an additional challenge to classification, often leading other systems to misidentify roads as building changes, a common problem in most conventional models.

However, ConMamba overcomes these challenges with its innovative CS-Hybridizer hybrid encoder. The encoder optimizes the feature extraction process by cleverly integrating the powerful spatial feature extraction capabilities of CNNs with the time-series processing advantages of SSMs through the FIM module, where CNNs are responsible for meticulously capturing spatial features while SSMs focus on parsing long-term dependencies in the time-series data, enhancing the model's sensitivity and adaptability to the dynamics of the environment. This fusion of techniques not only improves the model's recognition accuracy when dealing with complex scenes but is also particularly effective in coping with color interference and structurally complex environments, allowing ConMamba to reduce false positives in accurately

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

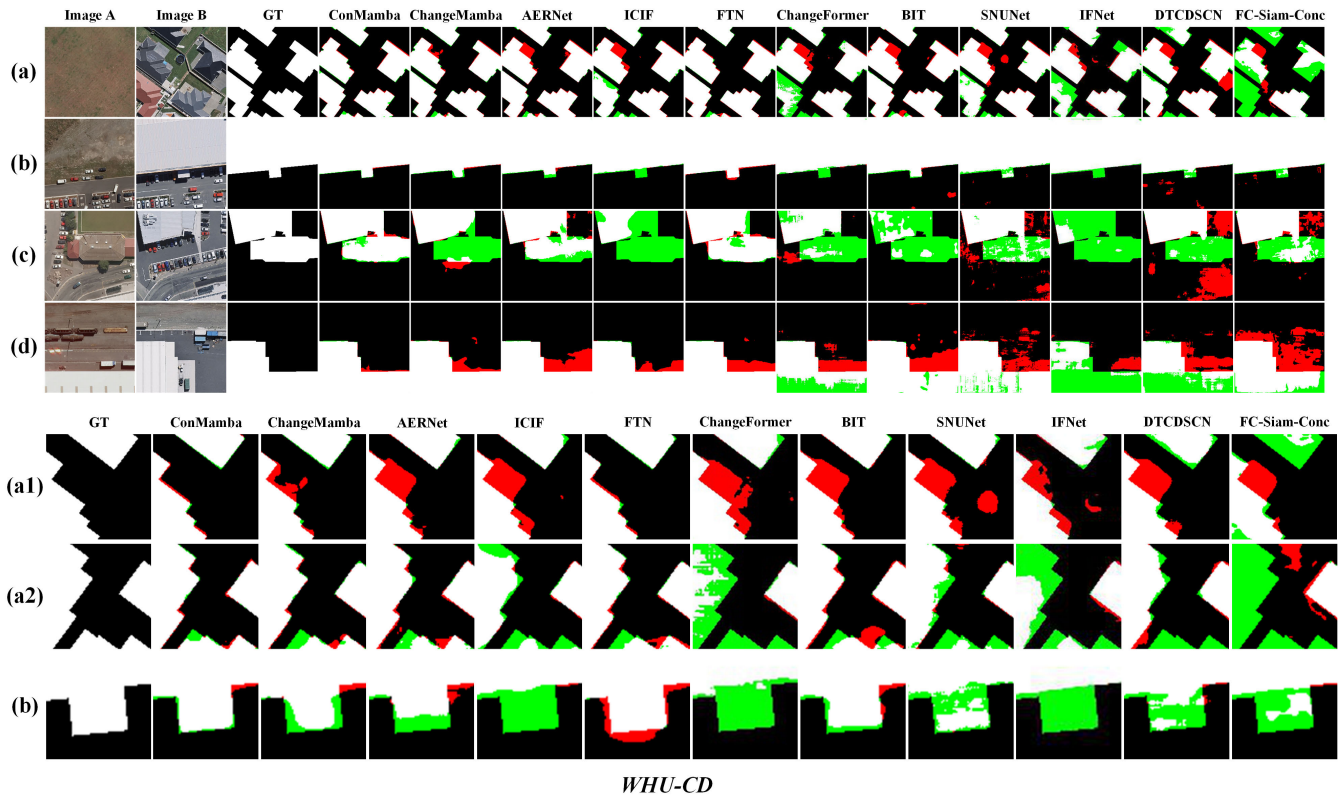


Fig. 6. Visualization results of ConMamba and comparison methods on the WHU-CD dataset. The upper images (a)–(d) overall detection results of dataset samples, with the lower images (a1) and (a2) presenting zoomed-in views of two local areas from the upper image (a), and the lower image (b) zoomed-in view of a local area from the upper image (b).

identifying real changes, demonstrating its superior performance in demanding CD tasks.

The data in Table II further validate the excellent performance of ConMamba on the WHU-CD dataset. The model achieves the highest scores in both integrated performance metrics, $F1$ and mIoU, ahead of the suboptimal model by 0.83% and 1.40%, respectively. In addition, ConMamba also achieved the optimal score in OA, showing its wide applicability. However, in Precision and Recall, ConMamba only ranked second, but this implies its potential and room for improvement in the continuous optimization process. These indicators not only highlight ConMamba's performance advantages in the field of CD but also point to the direction of future research and applications.

c) GZ-CD: The resolution of the GZ-CD dataset is slightly lower compared to the previous two datasets, thus placing higher demands on the feature recognition and noise suppression abilities of the model. To comprehensively evaluate the performance of our proposed ConMamba model on this dataset, we carefully selected four sets of representative samples for testing. As shown in Fig. 7, we show the visualized comparison results of ConMamba with other comparison models on the GZ-CD dataset, with local zoomed-in view displays of tiny targets in the first two sets of samples. It is obvious from the zoomed-in view that all the other comparison models except ConMamba fail to accurately identify the minutely changed regions in the lower-left corner of sample (a) and the lower-right corner of sample (b). In addition, ConMamba, although also affected by blurred features, misclassifies to a lesser extent.

TABLE II

RESULTS OF COMPARATIVE EXPERIMENTS ON THE WHU-CD DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUES HIGHLIGHTED IN RED AND THE SECOND HIGHEST IN BLUE

	WHU-CD				
Model	Precision	Recall	F1	mIoU	OA
FC-Siam-Conc	83.07	85.19	84.13	72.60	98.45
DTCDSCN	91.84	89.16	90.48	82.62	99.09
IFNet	**96.91**	73.20	83.41	71.53	98.83
SNUNet	91.34	85.53	88.34	79.11	98.91
BIT	88.71	86.27	87.47	77.73	98.81
ChangeFormer	88.50	85.33	86.88	76.81	98.76
FTN	93.09	91.24	92.16	85.45	**99.37**
ICIF-Net	92.83	88.70	90.77	83.09	99.13
AERNet	92.47	**91.89**	**92.18**	**85.49**	99.25
ChangeMamba	92.16	90.74	91.49	84.43	99.19
ConMamba	**94.79**	**91.28**	**93.01**	**86.89**	**99.39**

For samples (c) and (d), the presence of diachronic image color differences poses a challenge to the models' CD accuracy. Fig. 7 shows that when processing sample (c), most of the comparison models appear to misidentify the background as a change target, which manifests as multiple erroneous patches, which exposes their deficiencies in color sensitivity and background segmentation capabilities. For sample (d), the nontarget region in the lower left corner is highly susceptible to being misidentified as a changing region, while ConMamba effectively avoids such misclassifications,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

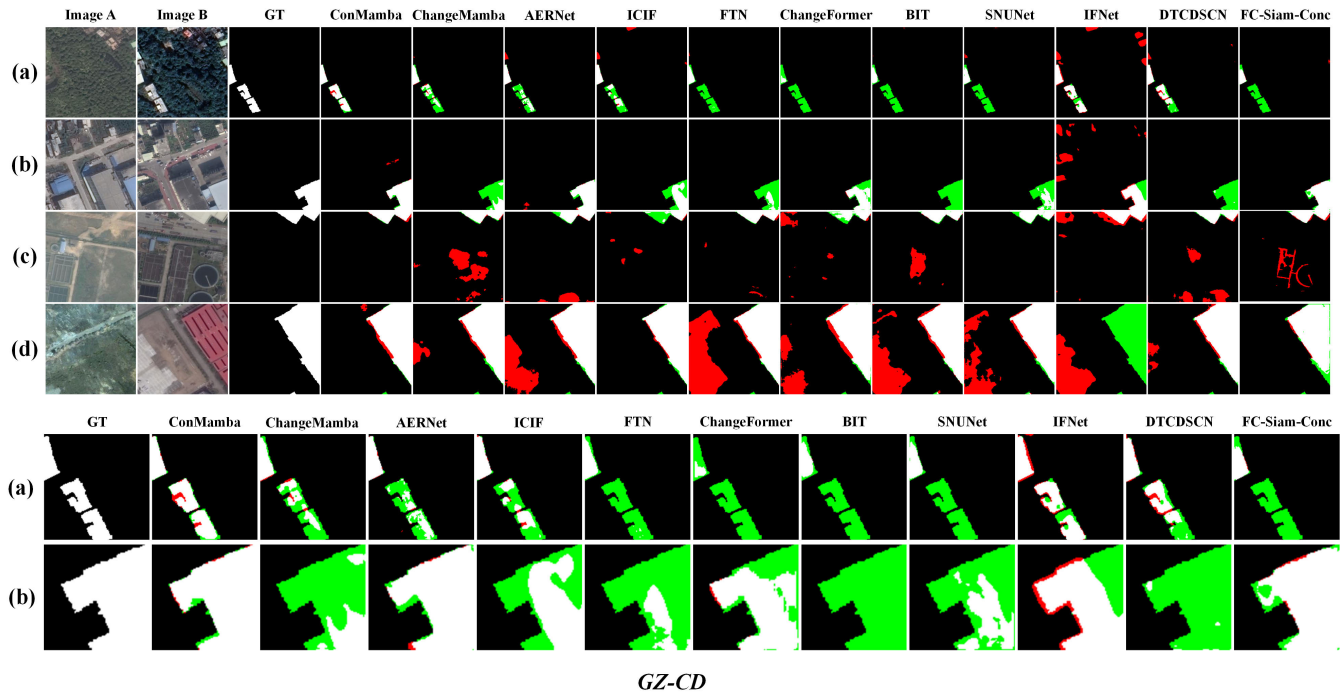


Fig. 7. Visualization results of ConMamba and comparison methods on the GZ-CD dataset. The upper images (a)–(d) illustrate the overall detection results of dataset samples, while the lower images (a) and (b) provide zoomed-in views of key areas from the upper images (a) and (b).

showing its significant advantages in detail processing and anti-interference ability. These results fully demonstrate the high performance and robustness of ConMamba in dealing with low-resolution and high-noise environments, highlighting its excellent performance in complex CD tasks.

The data results in Table III provide solid evidence of the excellent performance of the ConMamba model, especially in the key evaluation metrics of $F1$ score and mIoU, where the model achieves the highest score. This performance highlights ConMamba's superior ability to carefully balance recognition accuracy and recall, reflecting its strong efficiency in parsing complex change scenarios. Although it failed to lead in OA, ranking second, this did not diminish its advantage in the overall performance evaluation. These data emphasize ConMamba's leading position in the comprehensive evaluation metrics, proving its remarkable effectiveness and practical value in demanding CD tasks.

3) Comparative Analysis of Efficiency: In this section, we conduct an in-depth evaluation of the computational efficiency of various comparison methods based on two key indicators: parameter count (Params) and floating-point operations (FLOPs), as detailed in Table IV. The ConMamba model employs an innovative CNN and SSM hybrid architecture. While this design increases the network's depth and width, resulting in a parameter count of 34.18 M, placing it at a medium level among the comparison models, it still demonstrates significant advantages. Compared to CNN and Transformer hybrid architectures such as FTN (168.53 M) and ChangeFormer (41.03 M), ConMamba achieves a substantial reduction in parameter count while simultaneously exhibiting superior overall performance. Notably, compared to the standalone Mamba architecture of ChangeMamba, ConMamba reduces the parameter count by nearly 50 M and consistently outperforms ChangeMamba across multiple metrics on three remote sensing datasets. These results indicate that ConMamba not only maintains computational efficiency as a CNN and SSM hybrid model but also successfully avoids the computational bottleneck of quadratic complexity while effectively addressing the limitations of a single Mamba architecture in capturing fine-grained local details. This makes ConMamba a more robust and efficient solution for RSCD tasks.

In terms of FLOPs, ConMamba's higher computational demand is mainly due to the extensive use of its deep convolutional and transposed convolutional layers. Nevertheless, ConMamba's FLOPs are still significantly lower than

TABLE III

RESULTS OF COMPARATIVE EXPERIMENTS ON THE GZ-CD DATASET. ALL SCORES ARE EXPRESSED AS A PERCENTAGE (%), WITH THE HIGHEST VALUES HIGHLIGHTED IN RED AND THE SECOND HIGHEST IN BLUE

	GZ-CD				
Model	Precision	Recall	F1	mIoU	OA
FC-Siam-Conc	80.59	72.34	76.24	61.61	95.68
DTCDSCN	88.19	78.38	83.00	70.93	96.92
IFNet	**92.19**	74.08	82.15	69.71	96.92
SNUNet	87.55	80.05	83.64	71.87	97.00
BIT	82.40	78.18	80.23	66.99	96.31
ChangeFormer	84.59	65.23	73.66	58.30	95.53
FTN	86.99	**84.21**	**85.58**	**74.79**	**97.92**
ICIF-Net	**89.90**	80.76	85.09	74.05	97.29
AERNet	88.06	81.07	84.42	73.03	97.13
ChangeMamba	87.32	82.66	84.75	73.62	97.21
ConMamba	87.57	**83.74**	**85.78**	**75.10**	**97.35**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF THE EFFICIENCY OF THE METHODS

			LEVIR-CD		WHU-CD		GZ-CD	
	Params(M)	FLOPs(G)	F1	mIoU	F1	mIoU	F1	mIoU
FC-Siam-Conc	1.55	5.32	87.25	77.39	84.13	72.6	76.24	61.61
DTCDSCN	41.07	13.21	87.67	78.05	85.03	73.96	83.00	70.93
IFNet	50.71	82.35	89.32	80.70	83.41	71.53	82.15	69.71
SNUNet	12.03	54.88	88.13	78.83	85.26	74.31	84.25	72.79
BIT	3.55	10.59	89.31	80.68	84.9	73.75	80.23	66.99
ChangeFormer	41.03	202.87	90.40	82.48	81.82	69.24	73.66	58.3
FTN	168.53	45.00	91.01	83.51	92.16	85.45	85.58	74.79
ICIF-Net	25.83	25.27	91.18	83.85	90.77	83.09	85.09	74.05
AERNet	25.36	12.82	90.78	83.11	92.18	85.49	84.42	73.03
ChangeMamba	84.70	179.32	90.04	81.84	91.49	84.43	84.75	73.62
ConMamba	34.18	85.43	91.71	84.69	93.01	86.89	85.78	75.10

ChangeMamba's, showing its higher potential for real-world applications. It is worth emphasizing that VSS blocks significantly outperform traditional Transformer blocks in terms of floating-point efficiency. To concretely verify this advantage, we have carefully designed a comparison experiment by replacing the VSS blocks in the CS-Hybridizer architecture with Swin Transformer (ST) blocks and configuring the same number of block stacks in each of the three phases of the model [2, 2, 9]. This experiment aims to demonstrate the computational efficiency excellence of our developed CS-Hybridizer hybrid encoder, especially in comparison with the traditional CNN and Transformer hybrid architecture (CT-Hybridizer). This result further validates that ConMamba's design has the significant advantage of improving computational efficiency while maintaining a low floating-point computational burden, and is well suited for the RSCD task, which requires both computational efficiency and performance.

Table V summarizes the experimental results, showing the performance of the CS-Hybridizer encoder using a mixture of CNN blocks and VSS blocks compared to the CT-Hybridizer encoder using a mixture of CNN blocks and Swim Transformer blocks. The data shows that CS-Hybridizer significantly outperforms CT-Hybridizer in both Params and FLOPs, especially in FLOPs requirement, which is only half of CT-Hybridizer. In the performance tests on the three evaluation datasets, CS-Hybridizer outperforms CT-Hybridizer in all key performance metrics, especially on the LEVIR-CD dataset, where its $F1$ score and mIoU metrics are 0.55% and 0.87% higher, respectively. This result highlights the performance improvement of CS-Hybridizer by effectively exploiting the long-term dependency capture capability of SSM while maintaining computational efficiency. Compared to the Transformer, the efficiency of SSM mainly stems from the fact that it does not need to perform extensive self-attention computation on the whole sequence, which makes it more advantageous in scenarios dealing with complex temporal dynamics.

D. Ablation Experiments

In this study, we verify the validity of our contribution to the LEVIR-CD, WHU-CD, and GZ-CD datasets through ablation experiments. Three ablation variants of the ConMamba model are first explored in-depth: ConMamba-1, ConMamba-2, and ConMamba (Base). In ConMamba-1, we remove the CS-Hybridizer encoder and retain only the CNN branch. In ConMamba-2, the SIM module's expansion convolution is removed, leaving only the base single-stream convolution structure. In contrast, ConMamba (Base) removes both the CS-Hybridizer encoder and the inflated convolution in the SIM module, maintaining only the CNN branch and the base single-stream convolution structure.

When analyzing the data in Table VI, we observe a decrease in $F1$ and mIoU scores for ConMamba-1 compared to the full ConMamba model, highlighting the key role of the SSM branch in enhancing the model's ability to capture global information. Furthermore, ConMamba-2's performance significantly decreases on all three datasets after the removal of the inflated convolution and its interactive learning mechanism, which demonstrates the effectiveness of the inflated convolution in expanding the perceptual field and optimizing the model's ability to process contextually variant information. In contrast to ConMamba (Base), the performance of both ConMamba-1 and ConMamba-2 is significantly improved, which further validates the positive role of these modules in improving the overall performance of the model.

To explore in depth the contribution of the CNN branch and the SSM branch and their FIMs, we performed detailed internal ablation experiments defining three CS-Hybridizer encoder variants: CS-Hybridizer-1, CS-Hybridizer-2, and CS-Hybridizer-3. CS-Hybridizer-1 removes the FIM module in favor of a simple connectivity operation designed to evaluate the effect of the FIM module. CS-Hybridizer-2 retains only the CNN branch to analyze its standalone performance impact. CS-Hybridizer-3 relies exclusively on the SSM branch to test its standalone functionality.

Table VII shows the performance comparison between the variants and the full model. The significant performance degradation of CS-Hybridizer-1 highlights the crucial role of the FIM module in bridging the CNN and SSM branching features. Specifically, the FIM module effectively integrates the output feature information of CNNs and SSMs, bridging the inconsistencies in their feature representations and hierarchies. This result verifies the indispensability of the FIM module in realizing the efficient transfer and fusion of feature information and ensures that the complementarity between the CNN and SSM branches is fully utilized. The results of CS-Hybridizer-2 further reveal the insufficiency of the CNN branch alone in global information capture, which indirectly reflects the complementary role of the SSM branch. The performance of CS-Hybridizer-3 shows that the SSM branch alone has obvious limitations in processing complex remote sensing tasks, especially when dealing with feature-ambiguous datasets (e.g., GZ-CD). These experimental results not only verify that the synergistic effect of CNN and SSM branches is essential to improve the model performance in remote sensing image processing but also further emphasize the rationality and irreplaceability of the FIM module.

V. DISCUSSION

The RSCD task is of great significance in monitoring natural disasters, urban sprawl, and other surface changes and has become one of the core applications in the field of Earth observation. Although the current network architecture

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

TABLE V
PERFORMANCE COMPARISON OF CS-HYBRIDIZER AND CT-HYBRIDIZER

Module	Params(M)	FLOPs(G)	VSS	ST	LEVIR-CD		WHU-CD		GZ-CD	
					F1	mIoU	F1	mIoU	F1	mIoU
CS-Hybridizer	34.18	85.43	✓	×	91.71	84.69	93.01	86.89	85.78	75.10
CT-Hybridizer	42.95	168.73	×	✓	91.16	83.82	92.32	85.76	85.52	74.66

TABLE VI
ABLATION EXPERIMENTS ON CONMAMBA

Model	CS-Hybridizer	SIM	LEVIR-CD		WHU-CD		GZ-CD	
			F1	mIoU	F1	mIoU	F1	mIoU
ConMamba-1	×	✓	91.04	83.62	92.12	85.43	85.13	74.16
ConMamba-2	✓	×	91.32	84.01	92.51	86.17	85.34	74.45
ConMamba(Base)	×	×	90.43	82.55	91.92	85.09	84.27	72.88
ConMamba	✓	✓	91.71	84.69	93.01	86.89	85.78	75.10

TABLE VII
INTERNAL ABLATION EXPERIMENTS ON CS-HYBRIDIZER

Model	CNN	FIM	SSM	LEVIR-CD		WHU-CD		GZ-CD	
				F1	mIoU	F1	mIoU	F1	mIoU
CS-Hybridizer-1	✓	×	✓	91.34	84.06	92.46	86.01	85.39	74.55
CS-Hybridizer-2	✓	×	×	91.04	83.62	92.12	85.43	85.13	74.16
CS-Hybridizer-3	×	×	✓	90.85	83.28	91.77	84.94	84.02	72.52
CS-Hybridizer	✓	✓	✓	91.71	84.69	93.01	86.89	85.78	75.10

combining the CNN and Transformer dominates the RSCD task, recent studies have shown that SSM exhibits unique advantages in long-term time-series modeling and capturing complex dynamics, which makes it very suitable for the specific needs of the RSCD task. SSM is more efficient at modeling long-range dependencies than traditional Transformers, demonstrating significant computational advantages. However, relying on the Mamba architecture alone has the disadvantage of not being able to efficiently capture the fine-grained local information required for intensive prediction tasks such as RSCD (see Fig. 8).

To address this challenge, this study combines the strengths of the CNN and SSM, proposing an innovative architecture—ConMamba—to comprehensively capture multiscale feature information in remote sensing images. ConMamba represents the first exploration of a CNN and SSM hybrid model for RSCD tasks and has achieved remarkable success in feature extraction and dynamic CD. Utilizing an efficient FIM, we developed a seamless feature transmission mechanism between the CNN and SSM, resulting in the high-performance hybrid encoder, CS-Hybridizer. This encoder not only avoids the quadratic complexity issues associated with self-attention mechanisms but also compensates for the limitations of the standalone Mamba architecture in capturing local features, offering a more robust general encoding solution for RSCD tasks.

In extensive experimental validation, ConMamba significantly outperforms existing mainstream models on all three publicly available datasets, LEVIR-CD, WHU-CD, and GZ-CD, and especially outperforms other state-of-the-art methods, including ChangeMamba, in terms of performance metrics and computational efficiency. The experimental results

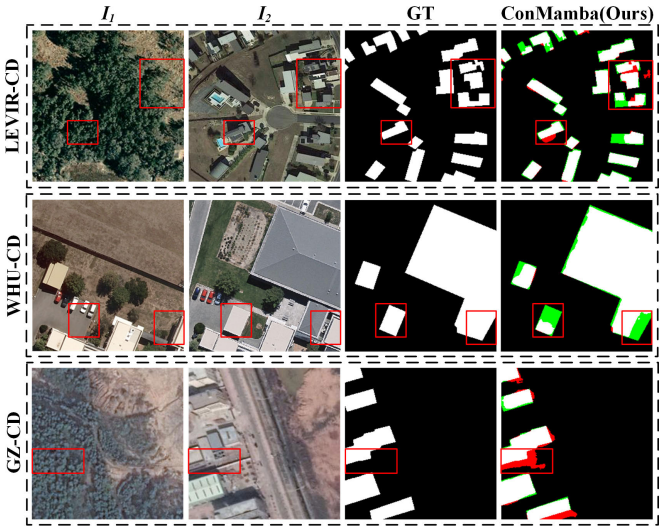


Fig. 8. Specific cases of prediction failure by ConMamba for three remotely sensed CD datasets. In the first two sets of samples, ConMamba successfully captured the overall changes by relying on the multiscale feature extraction capability of the CS-Hybridizer encoder, despite facing an extremely complex detection environment. However, due to the interference of the complex background, the model still suffered from a certain degree of misdetection and omission. In the third set of samples, ConMamba misidentified road shadows as buildings in the red box area due to the ambiguity caused by the low resolution of remote sensing images in the dataset. These results reveal the model's limitations in dealing with complex scenes and indicate potential directions for improvement.

show that ConMamba has successfully set a new benchmark in the RSCD task with its highly optimized architecture and excellent feature-capturing capability. This result not only validates the rationality and feasibility of our proposed CNN-SSM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

fusion architecture but also highlights its unique advantages in long-range dependency and fine-grained information extraction.

Although ConMamba has demonstrated excellent performance in experiments, it still has some limitations and needs to be further optimized. First, although ConMamba achieves a relative reduction in the number of parameters and floating-point operations compared with some hybrid CNN and Transformer models, its computational requirements are still high. Second, when dealing with image scenes with high blurring, the model fails to fully capture the subtle features, which may pose a potential risk to the accurate determination of surface changes.

To overcome the above limitations, future work will focus on improving the parameter efficiency and computational complexity of ConMamba to further reduce resource consumption and improve the applicability of the model in practical applications. In addition, we plan to enhance ConMamba's ability to capture features in high-ambiguity scenes to further improve its performance in complex remote sensing tasks. Through these improvements, we expect to make ConMamba a more efficient and robust RSCD model, adapting to a wider range of application scenarios and achieving higher accuracy.

In summary, the proposal of ConMamba is of great significance to the RSCD task. Its unique hybrid architecture and excellent experimental results provide new ideas for future RSCD methods and create a new technical path.

VI. Conclusion

In this work, we introduce a CNN and SSM-based hybrid model to the RSCD task for the first time and propose the baseline model ConMamba. To enable ConMamba to effectively model the multiscale features of various remote sensing images, we designed a streamlined FIM that integrates the CNN branch, composed of Residual Blocks, with the SSM branch, built with VSS blocks. This forms an efficient hybrid encoder, CS-Hybridizer, which is capable of fully capturing the global contextual information of input images with linear complexity. In addition, to further enhance the model's ability to parse and integrate complex contextual information, ConMamba introduces SIM in the decoding phase to enhance the understanding and refinement of complex contexts. Experiments on three public datasets show that ConMamba outperforms other mainstream architectures in the RSCD task, demonstrates significant competitive advantages and high adaptability, and provides a more robust technology path for the RSCD domain.

References

[1] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[2] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[3] Z. Lv, P. Zhong, W. Wang, Z. You, and N. Falco, "Multiscale attention network guided with change gradient image for land cover change detection using remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[4] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[5] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[6] O. Oktay et al., "Attention U-Net: Learning where to look for the pancreas," 2018, *arXiv:1804.03999*.

[7] J. M. J. Valanarasu, P. Oza, I. Hacihaliloglu, and V. M. Patel, "Medical transformer: Gated axial-attention for medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*, 2021, pp. 36–46.

[8] J. Ruan, S. Xiang, M. Xie, T. Liu, and Y. Fu, "MALUNet: A multi-attention and light-weight UNet for skin lesion segmentation," in *Proc. IEEE Int. Conf. Bioinf. Biomed. (BIBM)*, Las Vegas, NV, USA, Dec. 2022, pp. 1150–1156.

[9] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[10] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[11] X. Ge, J. Yu, and F. Kong, "Privacy-preserving Boolean range query with verifiability and forward security over spatio-textual data," *Inf. Sci.*, vol. 677, Aug. 2024, Art. no. 120929.

[12] Z. Dong, J. Li, and Z. Hua, "Transformer-based multi-attention hybrid networks for skin lesion segmentation," *Expert Syst. Appl.*, vol. 244, Jun. 2024, Art. no. 123016.

[13] Z. Dong, G. Yuan, Z. Hua, and J. Li, "Diffusion model-based text-guided enhancement network for medical image segmentation," *Expert Syst. Appl.*, vol. 249, Sep. 2024, Art. no. 123549.

[14] Z. Fu, J. Li, L. Ren, and Z. Chen, "SLDDNet: Stagewise short and long distance dependency network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3000319.

[15] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[16] X. Dong et al., "CSWin transformer: A general vision transformer backbone with cross-shaped windows," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12124–12134.

[17] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[18] K. Gregor, I. Danihelka, A. Mnih, C. Blundell, and D. Wierstra, "Deep autoregressive networks," in *Proc. Int. Conf. Mach. Learn.*, 2014, pp. 1242–1250.

[19] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[20] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," 2024, *arXiv:2404.03425*.

[21] Z. Zhang, A. Liu, I. Reid, R. Hartley, B. Zhuang, and H. Tang, "Motion mamba: Efficient and long sequence motion generation," 2024, *arXiv:2403.07487*.

[22] Z. Zheng and C. Wu, "U-shaped vision mamba for single image dehazing," 2024, *arXiv:2402.04139*.

[23] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[24] S. Li, H. Singh, and A. Grover, "Mamba-ND: Selective state space modeling for multi-dimensional data," 2024, *arXiv:2402.05892*.

[25] J. Liu et al., "Point mamba: A novel point cloud backbone based on state space model with octree-based ordering strategy," 2024, *arXiv:2403.06467*.

[26] C. Cortes and V. Vapnik, "Support-vector networks," *Mach. Learn.*, vol. 20, no. 3, pp. 273–297, Sep. 1995.

[27] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, pp. 5–32, Oct. 2001.

[28] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.

[29] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[30] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[31] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2021.

[32] Z. Zhou, M. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, vol. 11045, 2018, pp. 3–11.

[33] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[34] M. Noman et al., "Remote sensing change detection with transformers trained from scratch," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704214.

[35] M. Yin, Z. Chen, and C. Zhang, "A CNN-transformer network combining CBAM for change detection in high-resolution remote sensing images," *Remote Sens.*, vol. 15, no. 9, p. 2406, May 2023.

[36] X. Ye-tong, G. Xin-zhe, Z. Wei-qiang, Z. Yue, N. Hai-long, and L. Tao, "A remote sensing image change detection model based on cnn-transformer hybrid structure," *Comput. Modernization*, vol. 23, no. 7, p. 79, Jul. 2023.

[37] X. Song, Z. Hua, and J. Li, "Context spatial awareness remote sensing image change detection network based on graph and convolution interaction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3000316.

[38] Z. Li et al., "TFCNs: A CNN-transformer hybrid network for medical image segmentation," in *Proc. Int. Conf. Artif. Neural Netw.* Cham, Switzerland: Springer, 2022, pp. 781–792.

[39] M. Maaz et al., "EdgeNEXt: Efficiently amalgamated CNN-transformer architecture for mobile vision applications," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Oct. 2022, pp. 3–20.

[40] S. Chu, P. Li, M. Xia, H. Lin, M. Qian, and Y. Zhang, "DBFGAN: Dual branch feature guided aggregation network for remote sensing image," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 116, Feb. 2023, Art. no. 103141.

[41] T. Yan, Z. Wan, and P. Zhang, "Fully transformer network for change detection of remote sensing images," in *Proc. Asian Conf. Comput. Vis.*, 2022, pp. 1691–1708.

[42] E. Nguyen et al., "S4ND: Modeling images and videos as multidimensional signals using state spaces," 2022, *arXiv:2210.06583*.

[43] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[44] J. Ruan and S. Xiang, "VM-UNet: Vision mamba UNet for medical image segmentation," 2024, *arXiv:2402.02491*.

[45] J. Ma, F. Li, and B. Wang, "U-mamba: Enhancing long-range dependency for biomedical image segmentation," 2024, *arXiv:2401.04722*.

[46] Z. Xing, T. Ye, Y. Yang, G. Liu, and L. Zhu, "SegMamba: Long-range sequential modeling mamba for 3D medical image segmentation," 2024, *arXiv:2401.13560*.

[47] K. Chen, B. Chen, C. Liu, W. Li, Z. Zou, and Z. Shi, "RSMamba: Remote sensing image classification with state space model," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[48] J. Cao et al., "DO-Conv: Depthwise over-parameterized convolutional layer," *IEEE Trans. Image Process.*, vol. 31, pp. 3726–3736, 2022.

[49] A. Mao, M. Mohri, and Y. Zhong, "Cross-entropy loss functions: Theoretical analysis and applications," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 23803–23828.

[50] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.

[51] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[52] J. Zhang et al., "AERNet: An attention-guided edge refinement network and a dataset for remote sensing building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5617116.

[53] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[54] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.



Zhiwei Dong received the degree in computer science from the School of Information Science and Engineering, University of Jinan, Jinan, China, in 2022. He is currently pursuing the master's degree with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai, Shandong, China.

His research interests include medical image segmentation and target detection.



Genji Yuan received the bachelor's degree in electronic science and technology from Shandong Technology and Business University, Yantai, China, in 2017, and the master's degree in computer science and technology from Shandong Technology and Business University, in 2020, and the Ph.D. degree in computer science and technology from Qingdao University, Qingdao, China, in 2024.

His research interests include computer vision, pattern recognition, machine learning, data analysis, and intelligent transportation systems.



Zhen Hua received the B.S. and M.S. degrees in electrical automation from Taiyuan University of Technology, Taiyuan, China, in 1989 and 1992, respectively, and the Ph.D. degree in electronic information engineering from China University of Mining and Technology, Beijing, China, in 2008.

She is currently a Professor with Shandong Technology and Business University, Yantai, China. Her research interests include computer-aided geometric design, information visualization, virtual reality, and image processing.



Jinjiang Li received the B.S. and M.S. degrees in computer science from Taiyuan University of Technology, Taiyuan, China, in 2001 and 2004, respectively, and the Ph.D. degree in computer science from Shandong University, Jinan, China, in 2010.

From 2004 to 2006, he was an Assistant Research Fellow at the Institute of Computer Science and Technology of Peking University, Beijing, China. From 2012 to 2014, he was a Post-Doctoral Fellow at Tsinghua University, Beijing. He is currently a Professor with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai, China. His research interests include image processing, computer graphics, computer vision, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 14:27:47 UTC from IEEE Xplore. Restrictions apply.